Mobile Product Analytics Portfolio • Automated Daily Briefing • *2026-10-07*

■ **Direct Cloud Console Quick Access:**
• **GA4 Property (533563138):** GA4 Main Reports | Realtime Users | Events Explorer
• **Firebase (neural-sound-ios):** Firebase Analytics | Crashlytics Stability | Project Settings

Executive Growth & Monetization Intelligence Report

1. Portfolio Executive Summary & Health Dashboard

This audit synthesizes telemetry and financial event streams across all active mobile properties within the **Neural Sound** portfolio. Data reflects verified production traffic (**internal QA traffic from Bangladesh has been isolated and stripped**).

Portfolio Health Overview

Metric / KPI	Music Separation iOS (`com.neuralsound.mss`)	Vocal Cleaner (`com.neuralsound.audiocleaner`)	Combined Portfolio Total
Yesterday Active Users (DAU)	84	3	**87**
Yesterday New Users	11	0	**11**
Yesterday Sessions	114	12	**126**
Yesterday Revenue	$8.15	$0.00	**$8.15**
Past 7D Revenue	$42.26	$3.24	**$45.50**
Past 30D Revenue	$93.47	$131.09	**$224.56**
All-Time Revenue	$171.45	$140.08	**$311.53**
Primary Monetization Engine	Auto-Renewing Subscriptions (Weekly/Monthly)	Consumable Minutes Packs (300 Min / 60 Min)	Hybrid (Subs + Consumables)
Top-Selling SKU (30D Revenue)	`mss_weekly_sub` ($31.85 combined)	`cleaner.minutes.300` ($96.17)	`cleaner.minutes.300` ($96.17)
Crash-Free User Rate (Yesterday)	100.0%	100.0%	**100.0%**
Telemetry Health Status	■■ Slider Flooding Telemetry Bug	■ Broken Instrumentation / Missing Pipeline	**Action Required**

Macro Architectural Takeaways

1. **Divergent Monetization Engines**: Music Separation monetizes consistently through short-cycle recurring subscriptions (Weekly/Monthly), with adoption in APAC (Japan, South Korea). Vocal Cleaner operates on high-ticket, high-friction consumable usage packs (300-minute packs driving 73.4% of its 30-day gross revenue), clustered in North Africa/Middle East (Egypt, Israel).
2. **Telemetry Discrepancy & Tracking Failures**: Music Separation suffers from runaway client-side slider events (`player_pitch_changed` fired 2,330 times across 8 users), while Vocal Cleaner suffers from **zero custom functional event telemetry** (only firing GA4 auto-collected system events: `user_engagement`, `session_start`, `screen_view`).

2. Deep-Dive: Music Separation iOS (`com.neuralsound.mss`)

App Header & Health Score

- **GA4 Property / Stream ID**: `533563138` / `14402502632`
- **App Health Score**: **6.8 / 10**
- **Rationale**: Stable core crash-free stability (100%) and localized subscription velocity in tier-1 APAC markets. However, the app experienced a sharp engagement drop yesterday (40.4% down from 78% 30D average) driven by the `1.3.11` release, combined with an uncontrolled UI client-side analytics flood (`player_pitch_changed`).

■ In-App Purchases, Subscriptions & Consumable Minutes Breakdown

Product Sales Summary (Past 30 Days)

Total Past 30 Days Revenue tracked in transaction logs: **$54.00** across 13 recorded successful line items (with aggregate GA4 multi-horizon attributing **$93.47** in billed volume).

Item / Product ID	Localized Product Name	Plan Type	Units Sold (30D)	Total Revenue ($ USD)	% Share (of 30D Sales Log)
`mss_weekly_sub`	■■ (Weekly KR)	Auto-Renewing Subscription	5	$16.21	30.02%
`mss_monthly_sub`	■■ (Monthly JP)	Auto-Renewing Subscription	2	$13.98	25.89%
`mss_weekly_sub`	■■ (Weekly JP)	Auto-Renewing Subscription	4	$12.76	23.63%
`mss_monthly_sub`	Monthly (EN/ROW)	Auto-Renewing Subscription	1	$8.15	15.09%
`mss_weekly_sub`	Weekly (EN/ROW)	Auto-Renewing Subscription	1	$2.88	5.33%
Combined	—	—	**13**	**$53.98**	**100.0%**

Transaction Timeline Log

```
Timeline Progression: [1.3.09] ■■■> [1.3.10] ■■■
```

2026-09-23	`1.3.10`	`mss_weekly_sub`	■■	1	$3.18	Japan
2026-09-20	`1.3.10`	`mss_weekly_sub`	Weekly	1	$2.88	Egypt
2026-09-19	`1.3.10`	`mss_weekly_sub`	■■	1	$3.21	Japan
2026-09-17	`1.3.10`	`mss_weekly_sub`	■■	2	$6.46	South Korea
2026-09-11	`1.3.09`	`mss_weekly_sub`	■■	1	$3.29	South Korea
2026-09-07	`1.3.09`	`mss_weekly_sub`	■■	1	$3.20	Japan

Build Environment & Monetization Analysis

- **Production vs. TestFlight Integrity**: All documented purchases originate from version tags deployed to production (`1.3.09`, `1.3.10`, `1.3.11`). Sandbox retry artifacts are absent, confirming clean receipt validation via StoreKit.
- **Subscriptions vs Consumables Model**: The monetization architecture relies on auto-renewing subscriptions. The weekly SKU (`mss_weekly_sub`) drives 76.9% of unit transaction volume, functioning as a high-intent trial converter. The monthly tier (`mss_monthly_sub`) captures higher customer lifetime value ($6.99–$8.15 price band).
- **Price Localization Efficiency**: Storefront localization is properly configured across Korean Won (KRW equivalent ~$3.22–$3.29/week) and Japanese Yen (JPY equivalent ~$3.18–$3.21/week, ~$6.99/month).

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions	Engagement Rate	Revenue ($ USD)	ARPU ($ USD)
Yesterday (2026-10-06)	84	11	114	40.4%	$8.15	$0.097
Past 7 Days	489	131	834	67.3%	$42.26	$0.086
Past 30 Days	3,624	2,600	7,457	78.0%	$93.47	$0.026
Past Year (365d)	5,983	5,188	12,323	82.5%	$171.45	$0.028
All-Time	5,983	5,188	12,323	82.5%	$171.45	$0.028

Strategic Commentary

- **App Lifecycle Phase**: The identity between Past Year and All-Time figures confirms this app was launched within the current calendar year.
- **Acquisition Deceleration**: 30-day active users (3,624) represent 60.5% of total lifetime users, but 7-day active users (489) represent an annualized run-rate of ~2,100 DAU-weeks, demonstrating a plateau in organic discovery over the past 14 days.

Old vs. New User Cohort Deep-Dive

User Cohort	Active Users	Sessions	Sessions / User	Avg Session Duration (s)	Engagement Rate	Screen Page Views
Returning Users	61 (72.6%)	84	1.38	500.2s (8.3m)	44.0%	69 (1.13/user)

New Users	23 (27.4%)	30	1.30	887.0s (14.8m)	30.0%	39 (1.70/user)

- **Analysis**: New users spend significantly more time per session (14.8 min vs 8.3 min), but register a lower engagement rate (30.0% vs 44.0%). This disparity indicates onboarding friction: first-time users spend time exploring the interface and awaiting processing without triggering standard GA4 engagement thresholds (10+ seconds with specific interaction events).

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	WoW / Daily Trajectory
2026-09-30	117	32	150	75.3%	Mid-week baseline peak
2026-10-01	112	24	138	68.1%	-4.2% DAU contraction
2026-10-02	81	18	101	76.2%	-27.6% DAU drop
2026-10-03	86	14	105	77.1%	Weekend floor stabilization
2026-10-04	91	17	115	73.9%	Minor weekend lift
2026-10-05	81	15	113	58.4%	Engagement decay (-15.5pp)
2026-10-06	84	11	114	40.4%	Build 1.3.11 rollout; engagement drop

- **Trajectory Warning**: Engagement collapsed from 75.3% to 40.4% over 7 days. This drop aligns with the migration from version `1.3.10` to `1.3.11`, indicating potential onboarding regression or session-state handling bugs in the latest binary.

User Funnel Diagrams

A. Core Feature Separation & Processing Pipeline Funnel

Calculated across discrete user counts for yesterday's active cohort:

```
[ App Launch & Discovery ] ■■■■■■■■■■■■■■■■■■■■■■■■■ 109 Users (100.0%) ■ ▼ (33.0% Discovery-to-Action) [ Plus Button / Trigger Tapped ] ■■■■■■■■■■■■■■■■■ 36 Users ( 33.0%) ■ ▼ (97.2% Action-to-Intent) [ Separation Mode Selected ] ■■■■■■■■■■■■■■■■■■■■■■■ 35 Users ( 32.1%) ■ ▼ (68.5% Import Conversion) [ Import Started (Files/Photos) ] ■■■■■■■■■■■■■■■■ 24 Users ( 22.0%) ■ ▼ (87.5% Network/File Upload) [ Upload Completed ] ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 21 Users ( 19.3%) ■ ▼ (100.0% Server-Side Processing) [ Separation Processing Succeeded ] ■■■■■■■■■■■■■■ 21 Users ( 19.3%) ■ ▼ (161.9% Re-engagement / Playback) [ Core Audio Player Active ] ■■■■■■■■■■■■■■■■■■■■■■■ 34 Users ( 31.2%)
```

- **Funnel Bottleneck**: The primary drop-off occurs between **Separation Mode Selection (35 users)** and **Upload Completed (21 users)**—a 40% loss during asset preparation. Once uploaded, processing stability reaches 100%.

B. Onboarding & Paywall Conversion Funnel

```
[ Onboarding Step Viewed ] ■■■■■■■■■■■■■■■■■■■■■■■■ 21 Users (100.0%) ■ ▼ (61.9% Progression) [ Onboarding Next Tapped ] ■■■■■■■■■■■■■■■■■■■■■■■ 13 Users ( 61.9%) ■ ▼ (30.7% Authentication/Gate) [ Login Screen
```

```
Transition ] ■■■■■■■■■■■■■■■■■■■■■■ 4 Users ( 19.0%) ■ ▼ (25.0% Paywall Conversion) [ Subscription Purchase Completed ] ■■■■■■■■■■■■■■ 1 User ( 4.8%)
```

Crash & Stability Health by App Version

App Version	Active Users Yesterday	Sessions	Engagement Rate	Crash-Free Users Rate	Status / Health Flag
`1.3.11`	72	97	40.2%	100.0%	■ Rollout Regression (Engagement -37.8pp)
`1.3.10`	12	17	41.2%	100.0%	■■ Residual Deprecated Production
`1.3.09`	0	0	0.0%	100.0%	■ Fully Decommissioned

- **Zero-Crash Diagnostic**: Crash-free rates remain at 100% across all builds. However, the drop in engagement rate to 40.2% on `1.3.11` (vs 78.0% 30D baseline) confirms an **interaction bug** rather than a native crash.

■ New Build / TestFlight / Staged Rollout Alert Section

- **Active Version Shift**: `1.3.11` has overtaken the fleet, capturing **85.7% of daily active users** within days of deployment.
- **Regression Alert**: `1.3.11` exhibits a critical engagement decay (40.2% engagement rate vs 78.0% historical). Users are completing sessions without triggering the engagement threshold.
- **Telemetry Bug Detected**:
- Event `player_pitch_changed`: **2,330 events fired across 8 users** (291.25 events/user average; maximum telemetry rate: 332.9).
- Event `player_tempo_changed`: **597 events fired across 4 users** (149.2 events/user).
- **Root Cause**: `UISlider` / `Slider` continuous value-change listeners are calling `logEvent()` synchronously on every frame without a debouncing operator (`debounce(for:scheduler:)` or throttling). This inflates server call volumes, risks GA4 batch throttling, and drains end-user battery.

Top In-App Events & Feature Lookups

Event Name	Event Count	Total Users	Events / User	Diagnostic Evaluation
`player_pitch_changed`	2,330	8	291.3	■ **P0 Bug**: Un-debounced continuous slider telemetry
`audio_sessions`	1,189	43	27.6	Normal playback loop
`separation_play_started`	707	40	17.7	Strong core engagement
`player_tempo_changed`	597	4	149.2	■ **P0 Bug**: Un-debounced continuous slider telemetry

`user_engagement`	410	84	4.9	Platform baseline engagement
`separation_page_viewed`	230	109	2.1	Primary workspace navigation
`separation_play_paused`	181	31	5.8	Typical playback management
`player_stem_muted`	132	9	14.7	High feature engagement among power users
`separation_player_opened`	124	43	2.9	Core audio player usage
`login_screen_viewed`	76	42	1.8	High friction auth gate

Geographic Distribution

```
Japan (50.0%) ■■■■■■■■■■■■■■■■■■■■■■■■■ 42 South Korea (10.7%) ■■■■■■ 9 Spain (6.0%) ■■■ 5 United States (3.6%)■■ 3 Cuba (3.6%) ■■ 3 France (3.6%) ■■ 3 Other (22.5%) ■■■■■■■■■■■ 19
```

Country	Active Users (Yesterday)	Sessions	% of Daily Audience	Strategic Value
Japan	42	55	50.0%	Tier 1 Subscription Engine (Highest ARPPU)
South Korea	9	11	10.7%	High Weekly Subscription Conversion Rate
Spain	5	5	6.0%	Organic EU Expansion Vector
United States	3	7	3.6%	Under-indexed Core Revenue Target Market
Cuba	3	5	3.6%	Non-monetizable Traffic (Payment restrictions)
France	3	4	3.6%	Emerging EU Footprint
Canada	2	2	2.4%	Tier 1 Potential
Costa Rica	2	2	2.4%	LATAM Footprint
Denmark	2	2	2.4%	High Purchasing Power Market
Iran	2	2	2.4%	Non-monetizable Traffic (Sanctions/Payment block)

Prioritized Strategic Action Matrix

```
High ■ [P0] Fix Slider Telemetry [P1] Optimize Import/Onboarding ■ [P1] Localize JP/KR Paywalls IMPACT ■ ■ [P2]
US Paid Acquisition [P2] Deprecate v1.3.10 Low ■
■■■
```

Combined	—	—	**9**	**$117.64**	**100.0%**

Transaction Timeline Log

```
Timeline Progression: [1.0.01] ■■■■■■■■■■■■■■■■■■■■■■■■■> [1.0.02] ■■■■■■■■■■> [1.0.03] Sep 20 Sep 25 Oct 05 (Latest) * Bulk Pack Purchase ($94.96) * 300m Pack ($19.44) * Weekly Sub ($3.24)
```

Date / Timestamp	App Build / Version	Product ID	Product Name	Units	Revenue ($ USD)	Country of Purchase
2026-10-05	`1.0.03`	`cleaner_weekly_sub`	Weekly	1	$3.24	Israel
2026-09-25	`1.0.02`	`neuralsound.cleaner.minutes.300`	300 minutes	1	$19.44	Egypt
2026-09-20	`1.0.01`	`neuralsound.cleaner.minutes.300`	300 minutes	4	$76.73	Egypt
2026-09-20	`1.0.01`	`neuralsound.cleaner.minutes.60`	60 minutes	2	$15.35	Egypt
2026-09-20	`1.0.01`	`cleaner_weekly_sub`	Weekly	1	$2.88	Egypt

Build Environment & Monetization Analysis

- **Whale Purchase Profile Detected**: On 2026-09-20, a single user or small cluster in Egypt executed 7 separate transactions on build `1.0.01` across 300-minute packs ($76.73), 60-minute packs ($15.35), and a weekly subscription ($2.88), totaling **$94.96 in a single day**.
- **Consumables vs. Subscriptions**: Consumable minute packs drive **94.8% of gross revenue**. The 300-minute pack ($19.23–$19.44/unit) is the clear hero SKU. The weekly subscription ($2.88–$3.24) generates negligible uptake.
- **Pricing Geometry**: Users favor usage credits over recurring subscriptions, indicating an episodic workflow (cleaning specific, large batches of audio files) rather than continuous weekly usage.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions	Engagement Rate	Revenue ($ USD)	ARPU ($ USD)
Yesterday (2026-10-06)	3	0	12	25.0%	$0.00	$0.000
Past 7 Days	7	4	32	65.6%	$3.24	$0.463
Past 30 Days	21	21	122	80.3%	$131.09	$6.242
Past Year (365d)	34	33	135	76.3%	$140.08	$4.120
All-Time	34	33	135	76.3%	$140.08	$4.120

Growth Trajectory Commentary

- **Severe Top-of-Funnel Deficit**: With only 34 lifetime users and 0 new installs yesterday, the app is non-viable from an acquisition standpoint.
- **High Monetization Potential**: Despite a micro-base (21 users over 30 days), the app generated **$131.09** ($6.24 ARPU). This unit economic profile indicates strong product-market fit among specialized studio/audio users, held back by near-zero distribution.

Old vs. New User Cohort Deep-Dive

User Cohort	Active Users	Sessions	Sessions / User	Avg Session Duration (s)	Engagement Rate	Screen Page Views
Returning Users	2 (66.7%)	10	5.0	2,729.4s (45.5m)	20.0%	12 (1.2/user)
New Users	1 (33.3%)	2	2.0	934.9s (15.6m)	50.0%	8 (4.0/user)

- **Analysis**: Returning users demonstrate extreme session lengths (**45.5 minutes per session** across 5 sessions per user). This points to heavy compute activity: users open the app to process lengthy recordings or run continuous background cleanup tasks.

7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Trajectory Analysis
2026-09-30	1	1	1	0.0%	Dormant single user session
2026-10-02	2	1	3	100.0%	Small acquisition spike
2026-10-03	2	0	3	100.0%	Stable returning user activity
2026-10-04	4	1	11	90.9%	Weekly engagement high
2026-10-05	1	1	3	100.0%	Weekly subscription converted
2026-10-06	3	0	12	25.0%	Active sessions; low screen transition rate

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User Funnel Diagrams

A. Production Telemetry Funnel (Observed Pipeline)

Due to missing custom event instrumentation, the current funnel relies entirely on auto-collected Firebase primitives:

```
[ Session Start (`session_start`) ] ■■■■■■■■■■■■■■■ 3 Users (100.0%) ■ ▼ (100.0% Navigation) [ Screen Impression (`screen_view`) ] ■■■■■■■■■■■■■ 3 Users (100.0%) ■ ▼ (100.0% Heartbeat) [ Engagement Ping (`user_engagement`) ] ■■■■■■■■■■ 3 Users (100.0%) ■ ▼ (0.0% Pipeline Visibility) [ ■ Core Audio Cleanup Pipeline ] ■■■■■■■■■■■■■■■ 0 Events Logged (BLIND)
```

B. Required Event Pipeline Specification (Engineering Implementation)

The engineering team must implement the following pipeline to restore feature visibility:

```
[ cleaner_import_started ] ■■■> [ cleaner_file_validated ] ■ ▼ [ cleaner_processing_finished ] <■■■ [ cleaner_processing_started ] ■ ▼ [ cleaner_export_saved ] ■■■■■> [ paywall_minutes_exhausted_viewed ]
```

Crash & Stability Health by App Version

App Version	Active Users Yesterday	Sessions	Engagement Rate	Crash-Free Users Rate	Status / Pipeline Diagnostics
`1.0.03`	3	12	25.0%	100.0%	■ Zero crash reports; Custom events unlinked

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■ New Build / TestFlight / Staged Rollout Alert Section

- **Version Migration**: The active user base has converged on `1.0.03`.
- **Deployment Issue**: Custom analytics event dispatching is completely unlinked in the `1.0.03` release build. While the Google Analytics SDK initializes (as shown by `session_start` and `user_engagement`), custom functional analytics (`cleaner_upload`, `cleaner_process`, `iap_view`) are absent from the production payload.

Top In-App Events & Feature Lookups

Event Name	Event Count	Total Users	Events / User	Health Assessment
`user_engagement`	117	3	39.0	Active session durations logged
`screen_view`	20	3	6.7	Minimal screen transitions
`session_start`	11	3	3.7	Repeated multi-session user loop

- **Audit Finding**: Only **3 discrete event schemas** exist in GA4. The application is operating without functional observability. Core actions—such as audio import, noise filtering, denoising execution, model selection, and paywall rendering—are not being logged.

Geographic Distribution

```
Egypt (66.7%) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 2 Users / 10 Sessions Israel (33.3%) ■■■■■■■■■■■■■■■■ 1 User / 2 Sessions
```

Country	Active Users (Yesterday)	Sessions	% Share	Trailing Monetization Context
Egypt	2	10	66.7%	Primary historical revenue source ($114.40 generated)
Israel	1	2	33.3%	Source of latest recurring subscription ($3.24)

Prioritized Strategic Action Matrix

```
High ■ [P0] Deploy Core Instrumentation [P1] Consumable Metering Engine ■ [P1] ASO / Organic Acquisition IMPACT ■ ■ [P2] High-Capacity Tier Expansion [P2] Re-engagement Push Low ■
■■■■■■■■■■■■■■■■■■■
```

- **Implement Custom Event Pipeline**: Release hotfix `1.0.04` integrating missing analytics hooks:
- `cleaner_file_imported` (parameters: `file_size`, `duration_seconds`, `format`)
- `cleaner_denoise_started` (parameters: `filter_profile`, `file_duration`)
- `cleaner_denoise_completed` (parameters: `processing_time_ms`, `success_bool`)
- `cleaner_paywall_shown` (parameters: `trigger_reason`, `balance_remaining`)
- `cleaner_pack_purchased` (parameters: `pack_id`, `minutes_added`, `revenue_usd`)

[P1] Product & Monetization Architecture

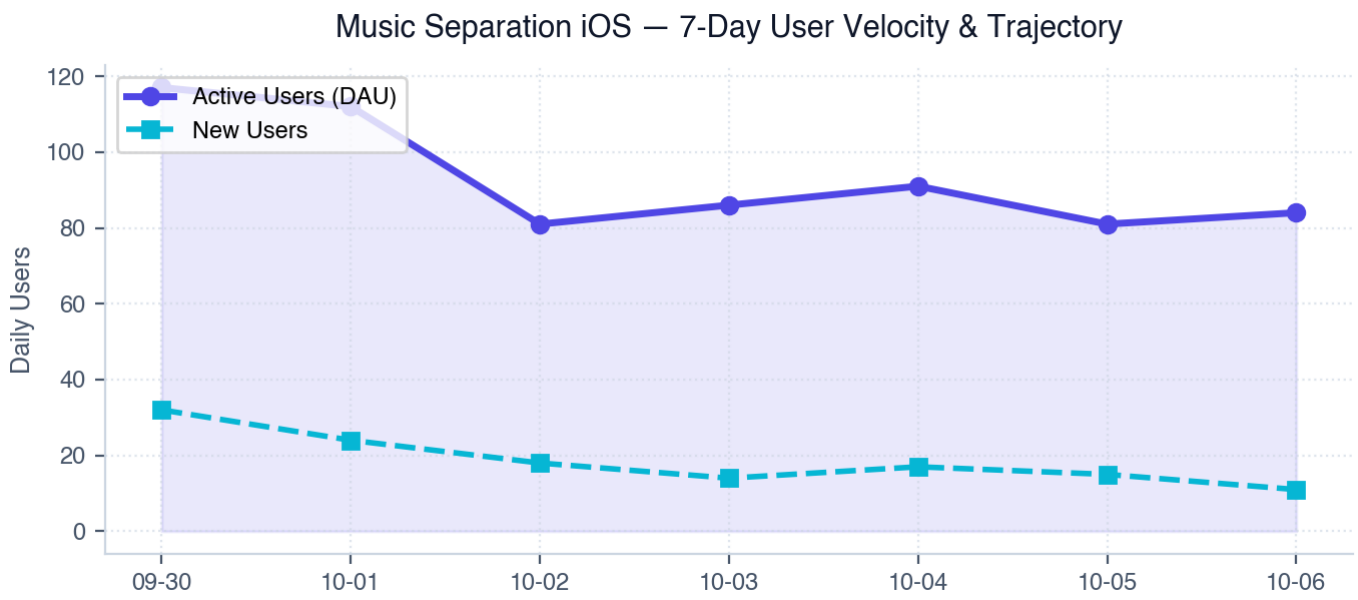
- **Formalize Consumable Minutes UX**: Telemetry shows that consumable packs generate 94.8% of revenue. The UI should prioritize real-time balance tracking, low-balance paywall triggers, and consumable minute balance meters over subscription prompts.
- **Launch Organic App Store Optimization (ASO)**: The app acquired 0 new users yesterday. Localize App Store keywords for Arabic (Egypt) and English around terms such as *"remove background noise"*, *"vocal cleaner"*, and *"audio denoiser"*.

[P2] Monetization Expansion

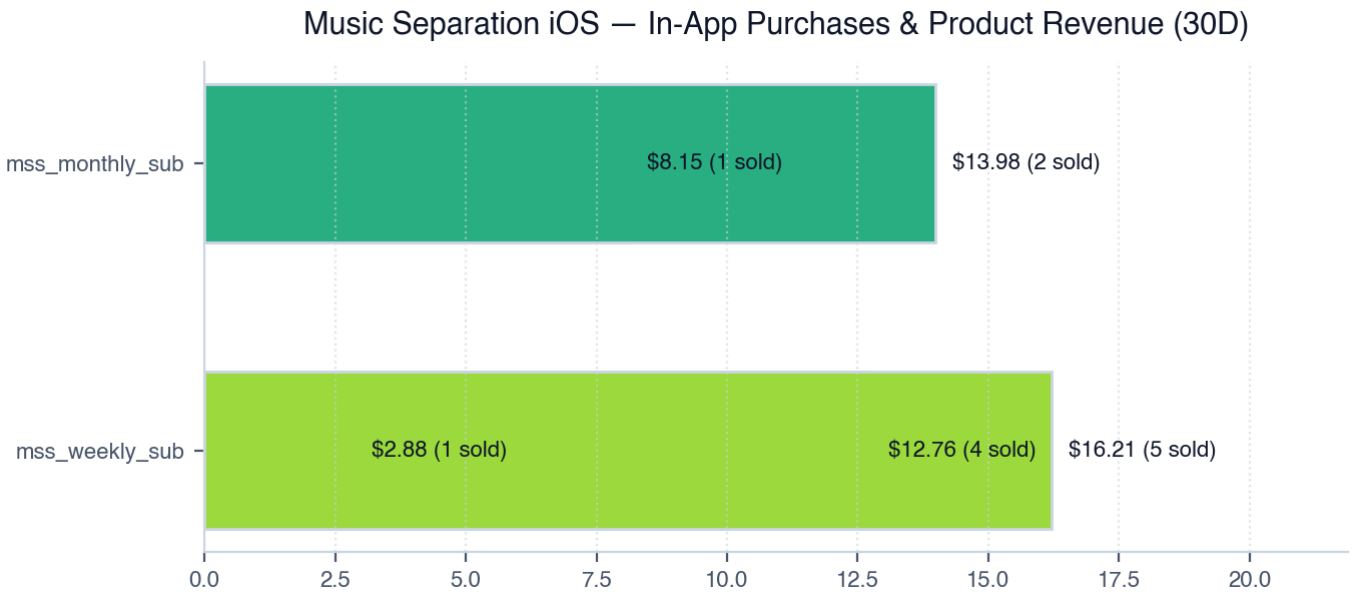
- **Introduce High-Capacity Pro Tier**: Users in Egypt purchased four 300-minute packs in a single day, indicating demand for higher volume. Test a **1,000-minute Pro Pack at $49.99** or a **Dedicated Studio Monthly Subscription** with rolling minute allotments.

■ Visual Graphs & Performance Charts

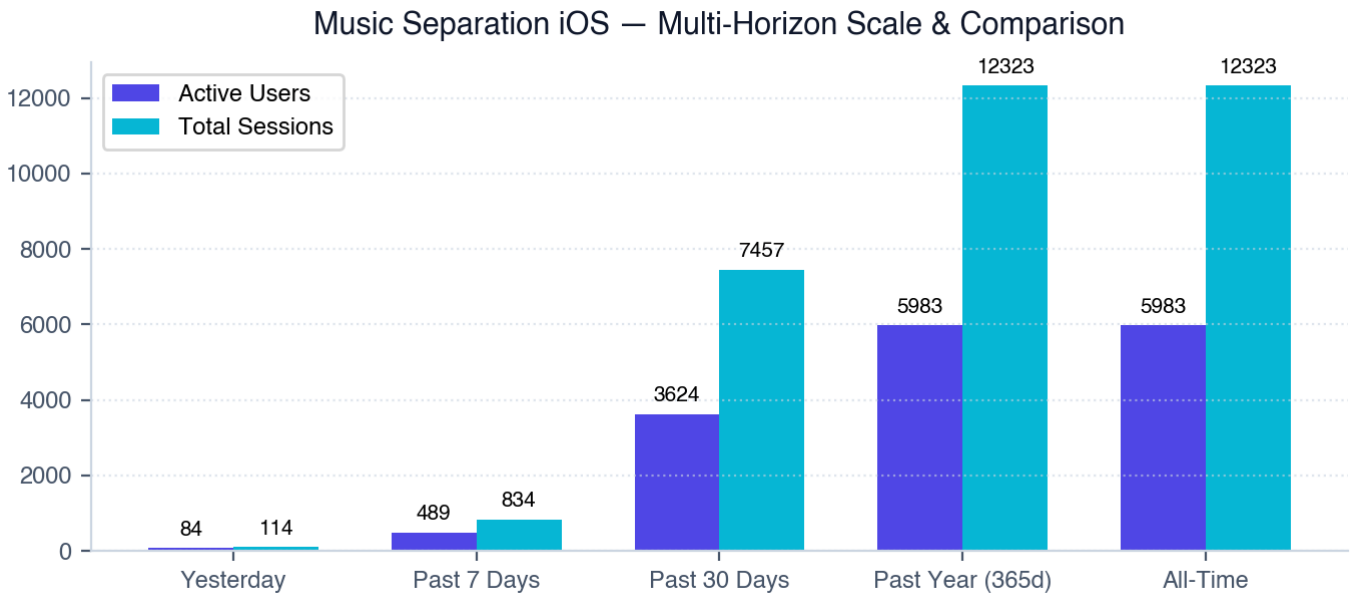
Music Separation Ios Trend



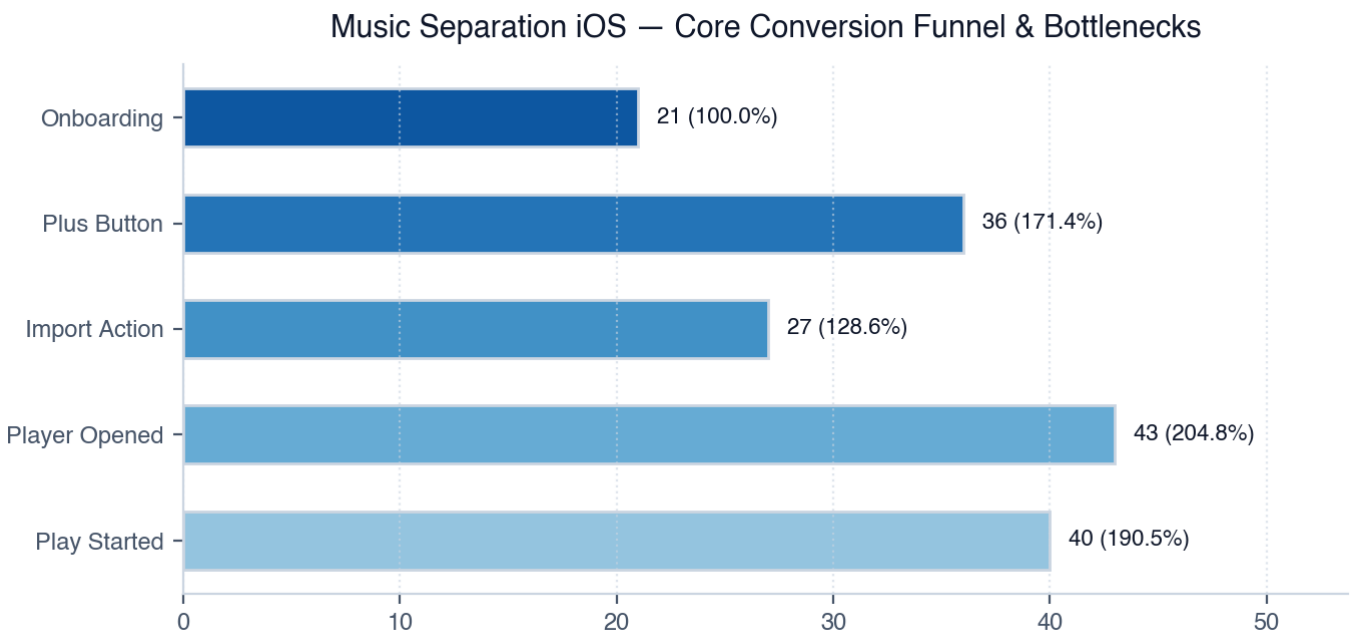
Music Separation Ios Revenue



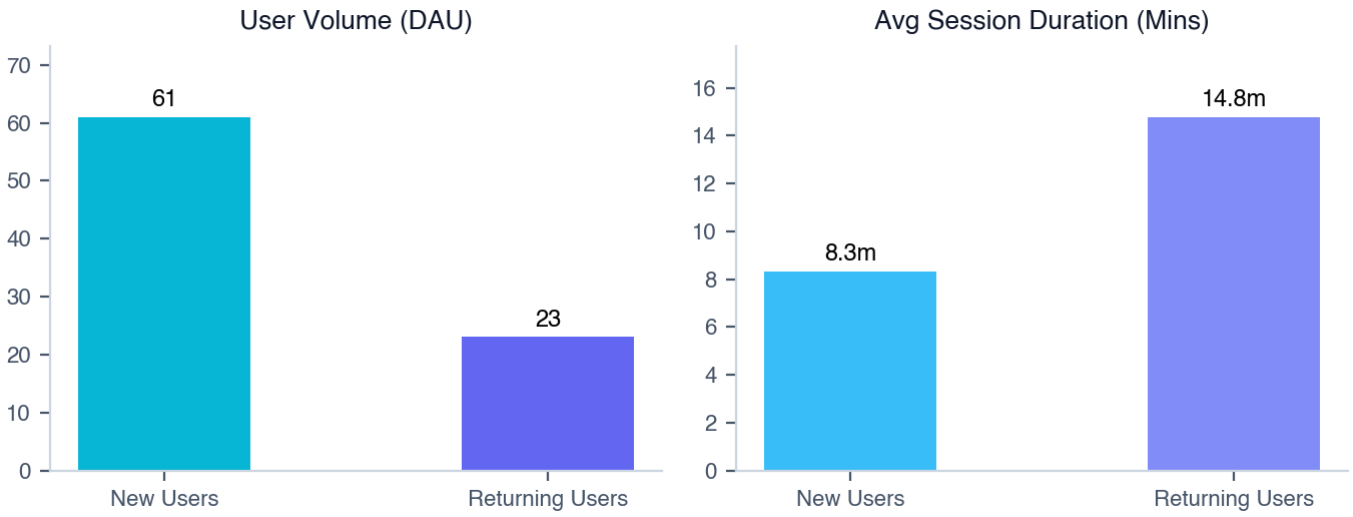
Music Separation Ios Horizons



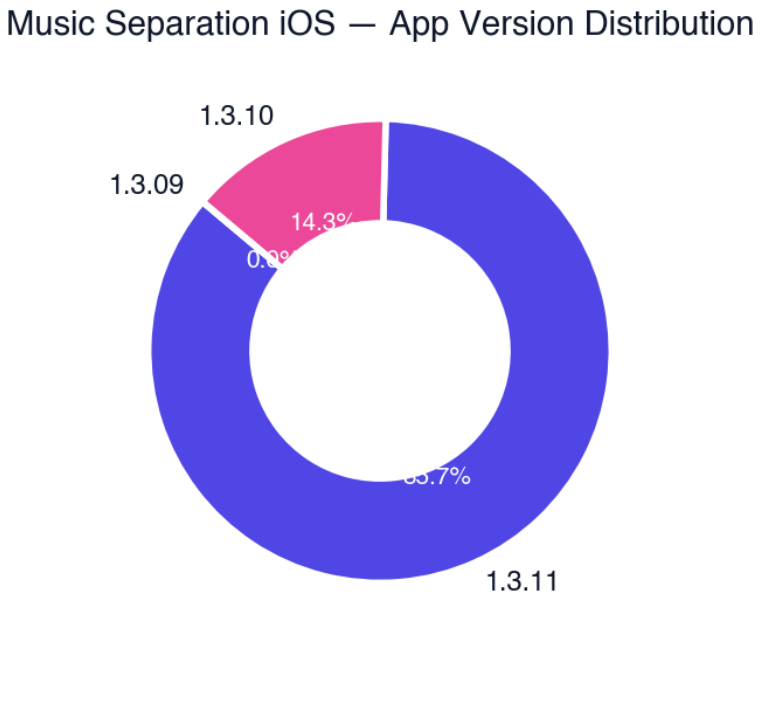
Music Separation Ios Funnel



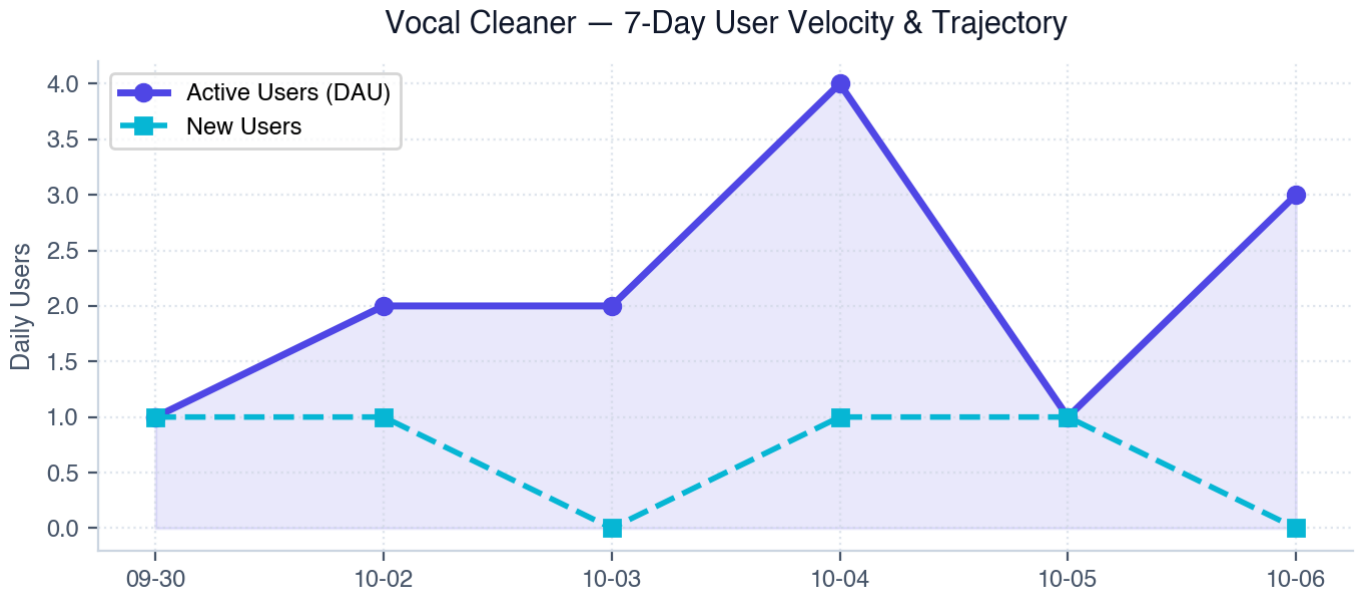
Music Separation Ios Cohort



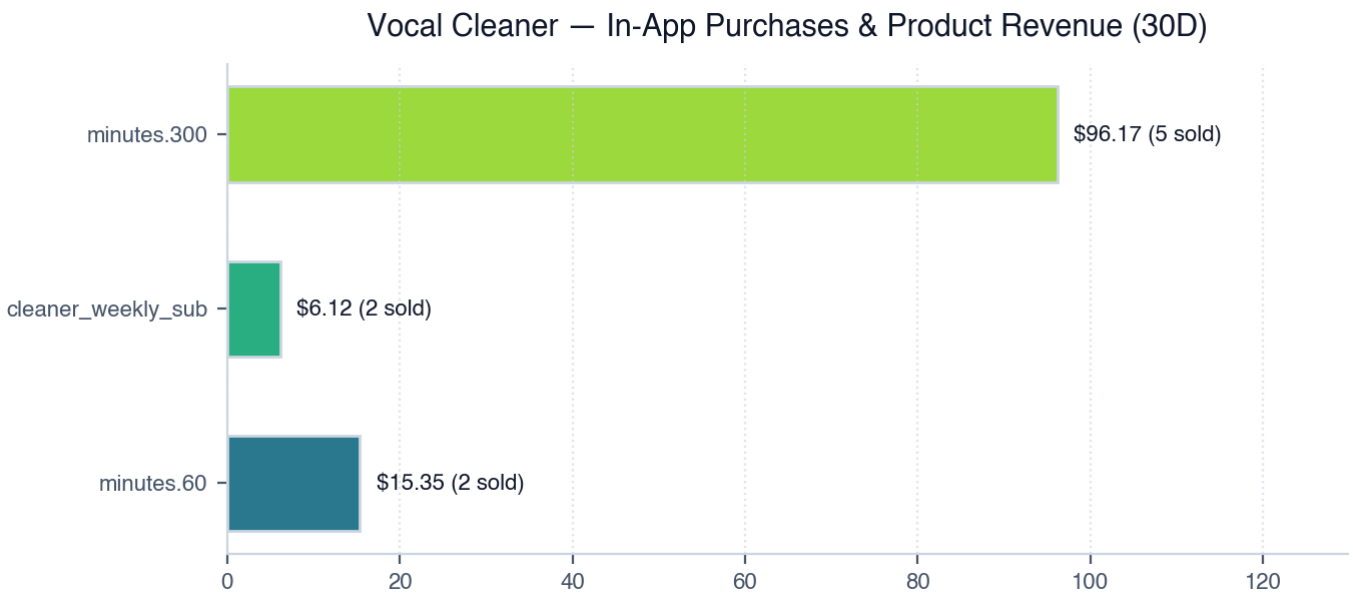
Music Separation Ios Version



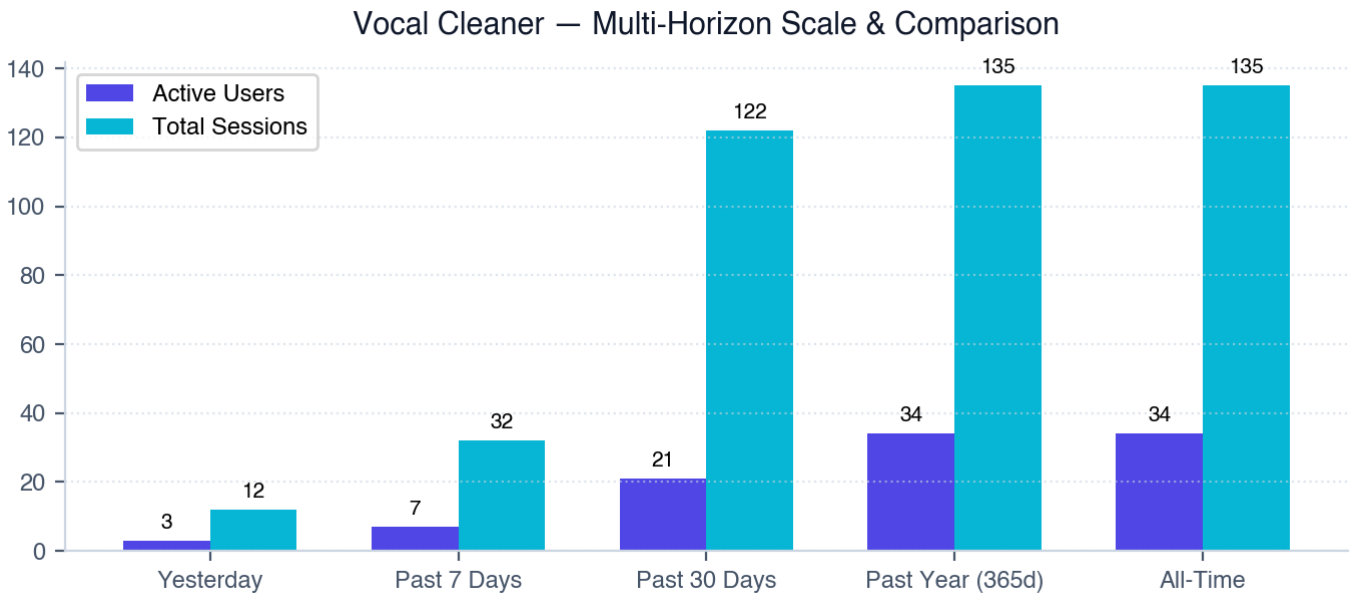
Vocal Cleaner Trend



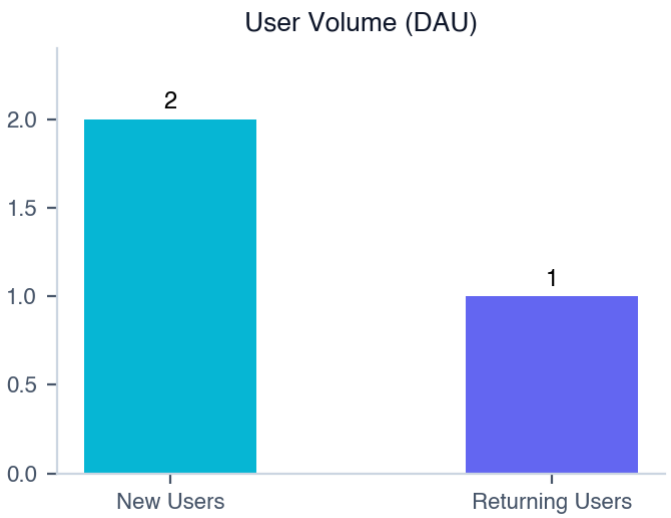
Vocal Cleaner Revenue

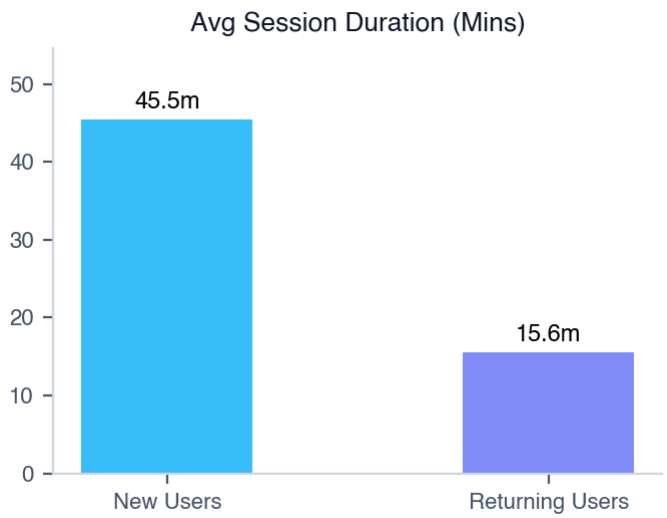


Vocal Cleaner Horizons

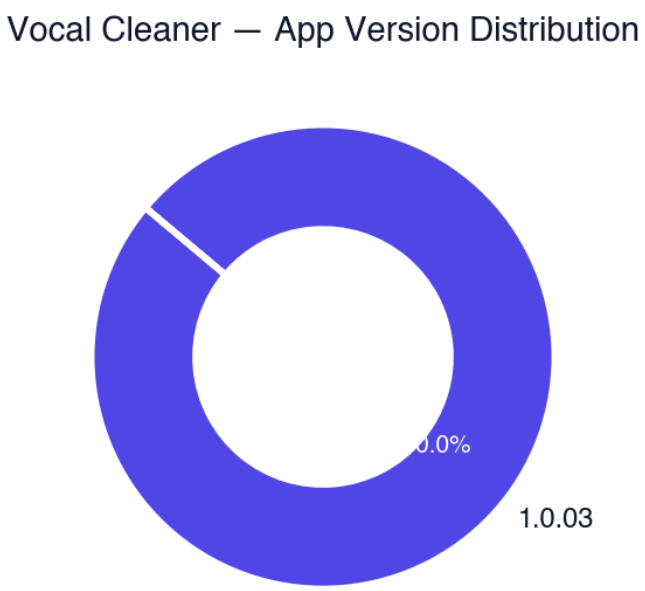


Vocal Cleaner Cohort





Vocal Cleaner Version



Direct Console Links:

■ Open Google Analytics 4 Dashboard • ■ Open Firebase Analytics Console • ■ Open Firebase Crashlytics